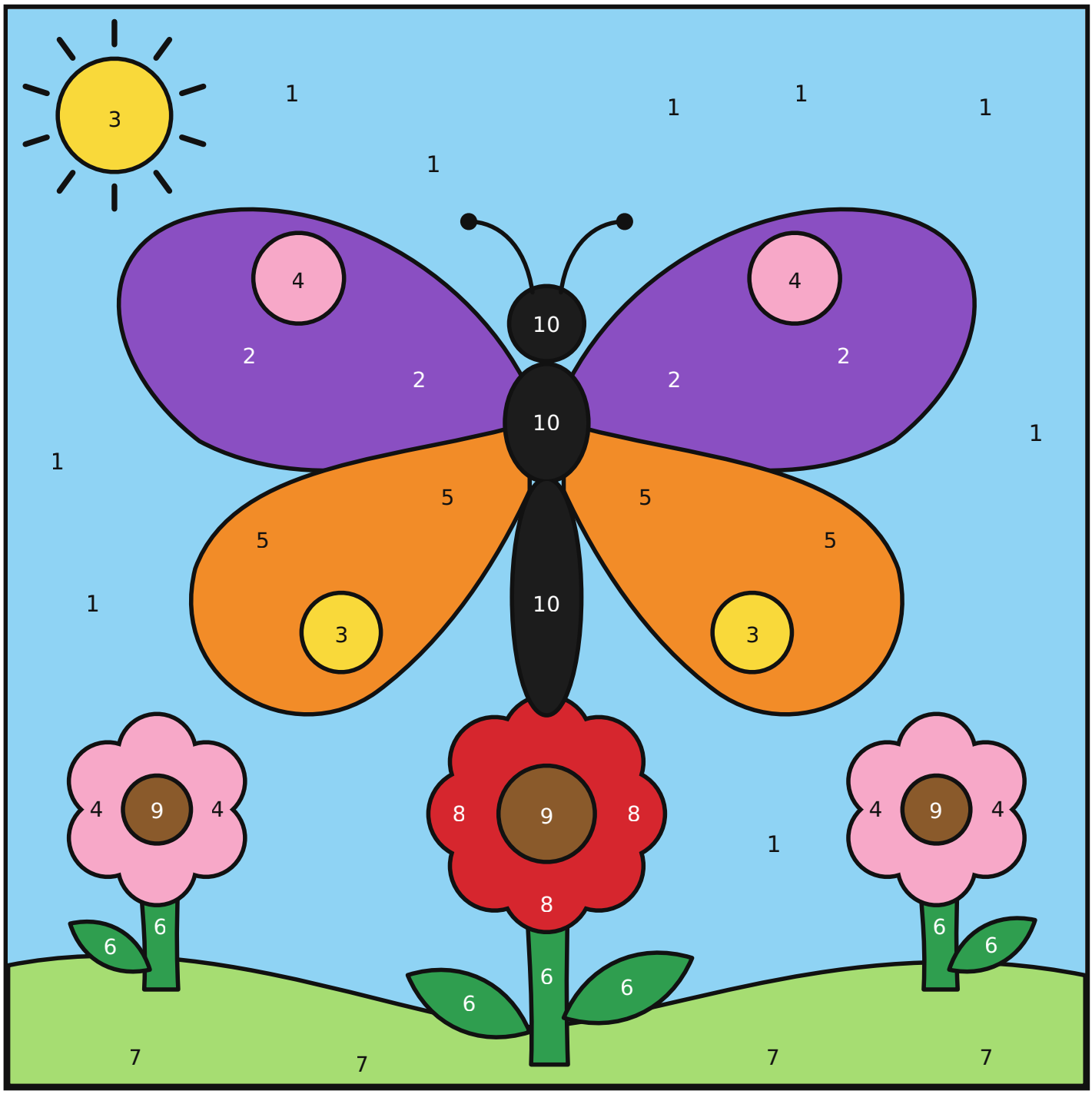
Name: **ANSWER KEY** _____ Date: _____

Exponents & Exponential Functions: Color By Number



Color Code:

6 Green $3x^5$	**9** Brown $x = 4$	**7** Light Green $\frac{1}{x^3}$	**5** Orange $12x^5$	**1** Light Blue x^7
8 Red 5	**4** Pink $4x^6$	**2** Purple x^5	**10** Black $y = 500(1.2)^x$	**3** Yellow x^{12}

Name: **ANSWER KEY** _____ Date: _____

Exponents & Exponential Functions: Color By Number

Directions: Solve each problem and show your work in the box. Your answer should match one of the answers in the Color Code on the coloring sheet. Find the matching answer, write the problem number on its blank, then write the color on the line. Color every part of the picture with that number. Watch out: the answers look alike!

1.

Simplify: $x^4 \cdot x^3$

add the exponents: 4 + 3 = 7

x^7

1. Color: **Light Blue**

2.

Simplify: $\frac{x^8}{x^3}$

subtract the exponents: 8 − 3 = 5

x^5

2. Color: **Purple**

3.

Simplify: $(x^3)^4$

multiply the exponents: 3 · 4 = 12

x^{12}

3. Color: **Yellow**

4.

Simplify: $(2x^3)^2$

square the 2 and multiply the exponents

$4x^6$

4. Color: **Pink**

5.

Simplify: $3x^2 \cdot 4x^3$

multiply coefficients: 3 · 4 = 12; add exponents: 2 + 3 = 5

$12x^5$

5. Color: **Orange**

6.

Simplify: $\frac{18x^7}{6x^2}$

divide 18 by 6 = 3; subtract exponents: 7 − 2 = 5

$3x^5$

6. Color: **Green**

7.

Write with a positive exponent: x^{-3}

a negative exponent means reciprocal

$\frac{1}{x^3}$

7. Color: **Light Green**

8.

Simplify: $5x^0$ $(x \neq 0)$

anything (not 0) to the power 0 is 1; 5 · 1 = 5

5

8. Color: **Red**

9.

Solve for x: $3^x = 81$

$81 = 3 \cdot 3 \cdot 3 \cdot 3 = 3^4$

x = 4

9. Color: **Brown**

10.

An account starts at $500 and grows 20% each year. Write the exponential function for the balance y after x years.

growth factor = 1 + 0.20 = 1.2

$y = 500(1.2)^x$

10. Color: **Black**